



CSE341: Microprocessor



Quiz - 4 Section: Marks: 15 Duration: 30 minutes
Name: ID:

Question-1: An 8086 microprocessor has the following segment addresses: **DS = 2000h**, **CS = 3000h**, **SS = 4000h** and while program **X** was executing, an **interrupt N** occurred. To handle this interrupt, the processor must jump to the **ISR** located at **386A9h**, which resides within the same code segment. Before fetching the ISR, the stack top offset was **3509h**. Some portions of memory are provided below:

Address	43509h	43508h	43507h	43506h	43505h	43504h
Data	57h	76h	32h	32h	70h	28h

Address	43503h	03AFh	03AEh	03ADh	03ACh	03ABh
Data	56h	30h	00h	86h	A9h	98h

- a. Deduce the value of CS and IP of program X. [3]
b. Deduce the memory addresses or locations of the IVT table where the CS and IP of Interrupt N is situated. [3]
c. Deduce the Interrupt Type N. [3]

a)

$(SS \times 10) + (SP - 3) = \text{Mem loc of CSH}$
 $(4000 \times 10) + (3509 - 3) = 43506 = \text{Mem loc of CSH}$

$[43506] \text{ CSH} = 32h$
 $[43505] \text{ CSL} = 70h$
 $[43504] \text{ IPH} = 28h$
 $[43503] \text{ IPL} = 56h$

$CS = 3270h$
 $IP = 2856h$

Stack diagram:
SP=6 → IPL
SP=5 → IPH
SP=4 → CSL
SP=3 → CSH
SP=2 → FlagL
SP=1 → FlagH
SP=3509 → X X

b)

IVT

IPL	03AC h
IPH	03AD h
CSL	03AE h
CSH	03AF h

$(CS \times 10) + IP = 386A9$
 $3000 \times 10 + IP = 386A9$
 $IP = 386A9 - 30000$
 $\therefore IP = 86A9h$ [IP of the starting address of ISR in IVT]

$\therefore \text{Memory Addresses}$
03AC h → IPL
03AD h → IPH
03AE h → CSL
03AF h → CSH

c)

Starting address of ISR = INT type $\times$ 4
03AC h = INT type $\times$ 4
2400 = INT type $\times$ 4

INT type = $\frac{240}{4}$
INT type = 235

b)

IVT

IPL	03AC h
IPH	03AD h
CSL	03AE h
CSH	03AF h

$(CS \times 10) + IP = 386A9$
 $3000 \times 10 + IP = 386A9$
 $IP = 386A9 - 30000$
 $\therefore IP = 86A9h$ [IP of the starting address of ISR in IVT]

$\therefore \text{Memory Addresses}$
03AC h → IPL
03AD h → IPH
03AE h → CSL
03AF h → CSH

c)

Starting address of ISR = INT type $\times$ 4
03AC h = INT type $\times$ 4
2400 = INT type $\times$ 4

INT type = $\frac{240}{4}$
INT type = 235

6. Data can now be transferred between Memory and I/O with the help of the DMA controller.
7. DMA controller sends interrupt signal to the CPU when finished with data transfer.
8. The DMA disables the DMA Acknowledgement signal and the CPU disables the Hold Acknowledgement signal taking back control of the system bus.